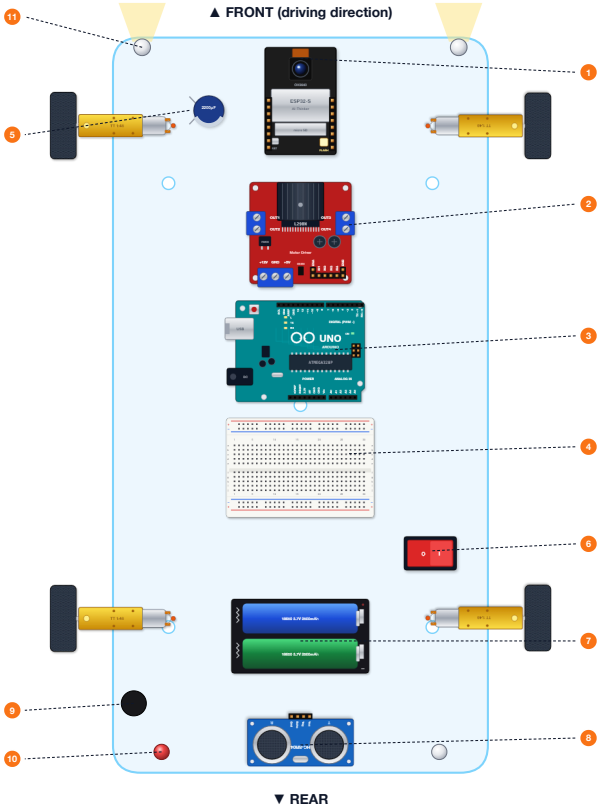
ROBOT CAR PROJECT

MODULE 03

03

Assembly Guide

Where every part goes on the chassis, how to mount it, and the mechanical safety check.



Name _____ Class / Group _____ Date _____

4WD · Arduino UNO · ESP32-CAM
Edition 2026

MODULE 03

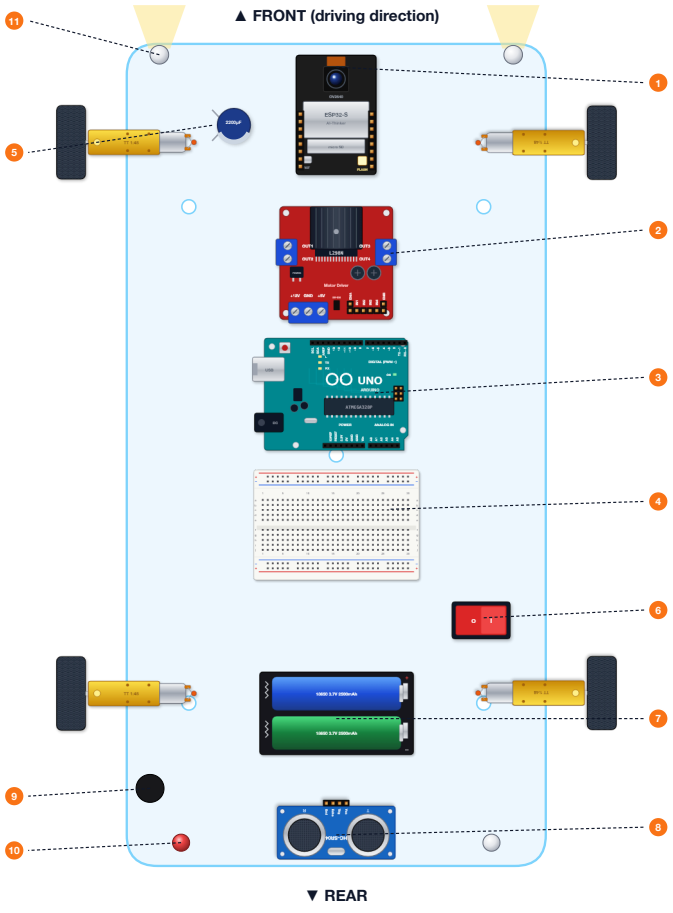
1 Mechanical assembly

A neat mechanical build makes wiring easier and the car more reliable. Motors and the battery go on the **bottom plate**; the electronics go on the **top plate**. Follow the layout below.

LEARNING OBJECTIVES · ABOUT 90 MIN

By the end of this module you can:

- Prepare motors with noise-suppression capacitors
- Place every part where it works best (camera, sensor, battery, antenna)
- Mount the rear sensor at the right height and angle
- Do a mechanical safety check before power-on



Top view: where each part goes (front at the top)

Layout legend

1. ESP32-CAM: front edge, lens forward, tilted up 5–10°
2. L298N motor driver: bottom plate, centre, heatsink free
3. Arduino UNO: top plate, centre
4. Breadboard: 5 V / GND rails, resistors, LEDs, buzzer
5. 2200 µF capacitor next to the ESP32-CAM 5V
6. ON/OFF switch: on the side, easy to reach
7. 2×18650 battery: low and towards the rear
8. HC-SR04: rear edge, facing back, 8–12 cm high
9. Buzzer: anywhere at the rear
10. Rear: red LED (A0) left, white reverse LED (A2) right
11. 2 white headlights (A4) at the front corners

WiFi antenna

Keep the ESP32-CAM antenna end (the zig-zag gold trace on the module) away from metal, motors and the L298N. Otherwise the WiFi range drops.

MODULE 03

2 Step by step

Step 1: Prepare the motors

1. Solder about 15 cm of wire to both tabs of each motor (red + black or two colours).
2. Solder a **100 nF ceramic capacitor** across the two tabs of each motor. This reduces electrical noise that disturbs the camera and WiFi.
3. Twist each motor's two wires together.
4. Mount the motors with the acrylic brackets so all **tabs face inwards**.

Step 2: Bottom plate

1. Fix the **L298N** in the centre, motor terminals facing the motors. Leave air around the heatsink.
2. Place the **battery holder** towards the rear. Mount the **switch** on the side.
3. Connect the motor wires to OUT1–OUT4 (Module 02, step 2).

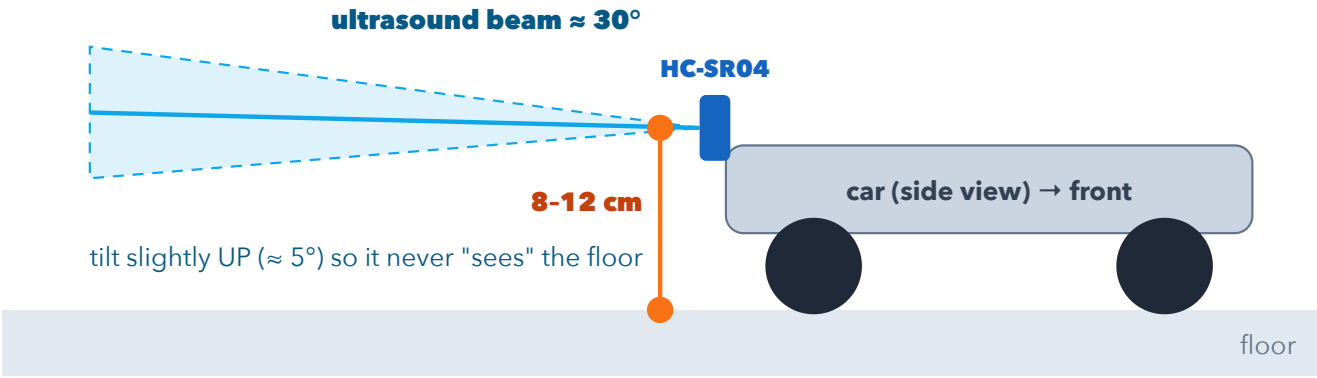
Step 3: Spacers and top plate

1. Screw in the brass spacers and mount the top plate.
2. Pass the control wires (ENA…ENB) and the power wires up through the centre opening.
3. Fix the **Arduino UNO** (nylon standoffs or foam tape) and stick the **breadboard** next to it.

Step 4: Front, the camera and the headlights

1. Mount the **ESP32-CAM at the very front**, camera looking forward, tilted up 5–10° (3D-printed holder, L-bracket or foam tape).
2. Make sure the camera ribbon cable is fully inserted and the black latch is closed.
3. Fix the **two white headlights** at the front corners, pointing forward (5 mm LED holders or hot glue). Do not cover the camera lens.
4. Run the headlight wires back to the breadboard along the side of the plate (long leg = + towards its 220 Ω resistor).

Step 5: Rear, sensor, buzzer and LEDs



Side view: correct sensor height and angle

1. Mount the **HC-SR04 on the rear edge, facing straight back**, 8–12 cm above the floor, tilted up slightly.
2. Nothing (wires, screws, wheels) may stick out in front of its two "eyes".
3. Fix the red LED at the left rear corner (like a brake light), the white reverse LED at the right rear corner, and the buzzer at the rear.

> **Sensor too low or pointing down?**
> It will detect the floor and beep, or block reversing, all the time. Raise it and tilt it up.

Step 6: Cable management

- Keep the power wires short. Use zip ties to keep wires away from the wheels.
- Route signal wires away from motor wires where possible.
- Battery low and centred or slightly to the rear: a skid-steer car turns better that way.

Mechanical check

- [ ] All 4 wheels turn freely by hand
- [ ] No wire can touch a wheel
- [ ] Battery holder is firmly fixed
- [ ] Camera faces forward and is tilted up a little
- [ ] Sensor faces back, 8–12 cm high, nothing in front of it
- [ ] Headlights at the front, pointing forward
- [ ] Switch is reachable

Built by (student) _____

Checked by (teacher) _____